

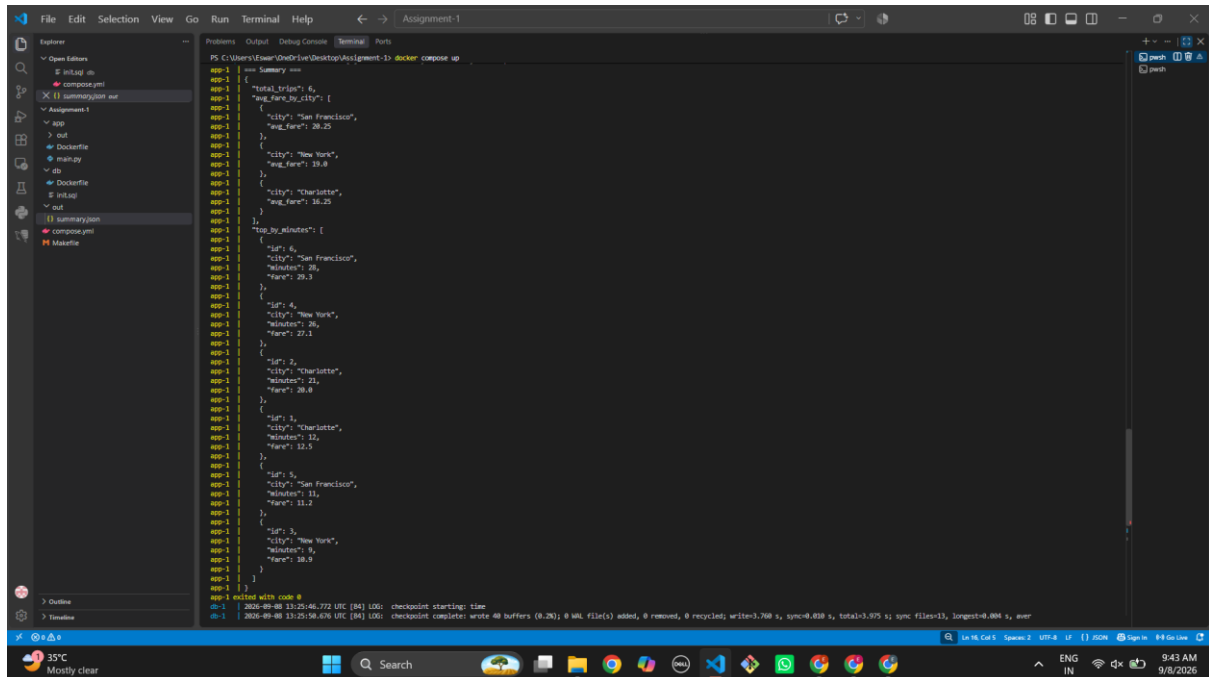
Commands used to build and run

docker compose up

stop containers:

docker compose down

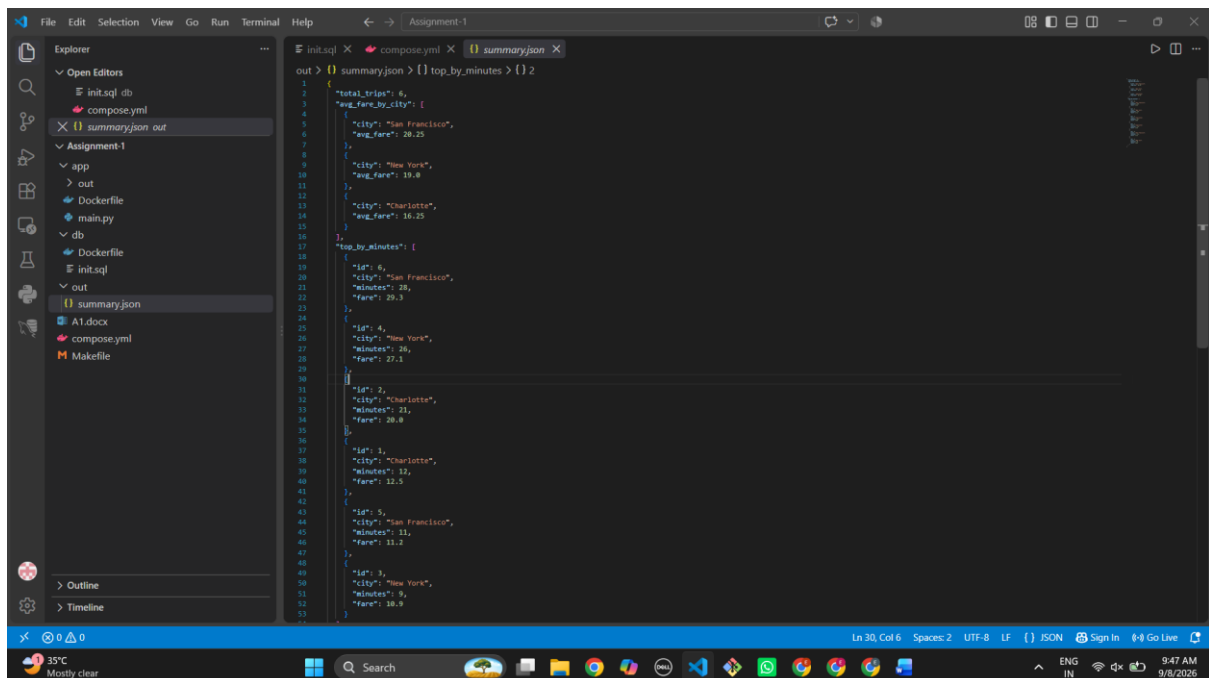
Console output



The screenshot shows a VS Code terminal window with the command `docker compose up` executed. The output displays the startup of several services: `db`, `app`, `out`, `summary`, and `compose`. The `summary` service outputs a JSON array of flight data, including total trips, average fare by city, and top flights by minutes. The `compose` service outputs a log message indicating the checkpoint starting time.

```
PS C:\Users\Ewan\OneDrive\Desktop\Assignment-1> docker compose up
app-1 | Summary
app-1 | {
app-1 |   "total_trips": 6,
app-1 |   "avg_fare_by_city": {
app-1 |     "city": "San Francisco",
app-1 |     "avg_fare": 28.25
app-1 |   },
app-1 |   "city": "New York",
app-1 |   "avg_fare": 19.0
app-1 | },
app-1 |   "city": "Charlotte",
app-1 |   "avg_fare": 16.25
app-1 | },
app-1 |   "top_by_minutes": {
app-1 |     "id": 6,
app-1 |     "city": "San Francisco",
app-1 |     "minutes": 28,
app-1 |     "fare": 29.3
app-1 |   },
app-1 |     "id": 4,
app-1 |     "city": "New York",
app-1 |     "minutes": 26,
app-1 |     "fare": 27.1
app-1 |   },
app-1 |     "id": 2,
app-1 |     "city": "Charlotte",
app-1 |     "minutes": 21,
app-1 |     "fare": 20.0
app-1 |   },
app-1 |     "id": 1,
app-1 |     "city": "Charlotte",
app-1 |     "minutes": 12,
app-1 |     "fare": 12.5
app-1 |   },
app-1 |     "id": 5,
app-1 |     "city": "San Francisco",
app-1 |     "minutes": 11,
app-1 |     "fare": 11.2
app-1 |   },
app-1 |     "id": 3,
app-1 |     "city": "New York",
app-1 |     "minutes": 9,
app-1 |     "fare": 10.9
app-1 |   }
app-1 | }
app-1 | }
db-1 | 2026-09-08 11:25:46.772 UTC [84] LOG: checkpoint starting: time
db-1 | 2026-09-08 11:25:58.626 UTC [84] LOG: checkpoint complete: write 48 buffers (0.2%); 0 WAL file(s) added, 0 removed, 0 recycled; write=3.708 s, sync=0.038 s, total=3.975 s; sync files=11, longest=0.004 s, ev
```

Summary.json



The screenshot shows the VS Code editor with the `summary.json` file open. The file contains a JSON array of flight data, including total trips, average fare by city, and top flights by minutes. The data is structured as follows:

```
{
  "total_trips": 6,
  "avg_fare_by_city": {
    "city": "San Francisco",
    "avg_fare": 28.25
  },
  "city": "New York",
  "avg_fare": 19.0
},
  "city": "Charlotte",
  "avg_fare": 16.25
},
  "top_by_minutes": {
    "id": 6,
    "city": "San Francisco",
    "minutes": 28,
    "fare": 29.3
  },
    "id": 4,
    "city": "New York",
    "minutes": 26,
    "fare": 27.1
  },
    "id": 2,
    "city": "Charlotte",
    "minutes": 21,
    "fare": 20.0
  },
    "id": 1,
    "city": "Charlotte",
    "minutes": 12,
    "fare": 12.5
  },
    "id": 5,
    "city": "San Francisco",
    "minutes": 11,
    "fare": 11.2
  },
    "id": 3,
    "city": "New York",
    "minutes": 9,
    "fare": 10.9
  }
}
```

Refelction

In this assignment, I learned how to orchestrate multi-container setups using Docker Compose and internal service DNS. I also gained practical experience handling database connection retries during container startup. For a production environment, I would secure the database credentials in a `.env` file and use multi-stage builds to keep the Docker images lightweight.